

Solution Requirements

Date	27 June 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Not applicable — the application does not require user login; it is a publicly accessible, single-session evaluation tool	Low
2	Authorization levels	Not applicable — no user roles or permission tiers exist; all users have identical access to the analysis tool	Low

3	External interfaces	The system integrates with the OpenAI Whisper model (local inference), Sentence-BERT (local inference via HuggingFace), and the Google Gemini API (external service) for AI-generated feedback	High
4	Transactions processing	Not applicable — the application does not process payments or financial transactions	Low
5	Reporting	The system generates a downloadable PDF report per analysis session, containing the transcript, semantic and fluency scores, AI-generated feedback, and a waveform visualization	High
6	Business rules, etc.	The system computes a final score using a fixed weighted formula: 60% semantic understanding + 40% fluency score, with results mapped to qualitative labels (Strong/Moderate/Poor Understanding)	High
7	Compliance to laws or regulations	The application does not store personal user data; uploaded audio is processed transiently and not retained beyond the session, minimizing privacy concerns	Medium
8	<i>Other</i>	The system must dynamically support multiple concepts via a structured reference dataset, allowing new concepts to be added without code changes	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	Describes end-to-end processing time for a single audio analysis request	Complete analysis (transcription + scoring + AI feedback) within 20-30 seconds on free-tier CPU hosting
2	Scalability	Describes how the system handles multiple/concurrent users	Currently supports single-user sequential processing; designed to be upgradable to concurrent request handling with paid hosting tiers
3	Security & Data Privacy	Describes protection of sensitive data, primarily the Gemini API key	API key stored securely via environment variables/Hugging Face Secrets, never hardcoded or committed to version control
4	Reliability & Availability	Describes uptime expectations given free-tier hosting constraints	Application available on-demand; free-tier hosting may sleep after 48 hours of inactivity, requiring ~30-60 sec to resume on next access
5	Usability & Accessibility	Describes user experience standards for the interface	Simple, single-page interface requiring no technical knowledge; clear visual feedback (scores, waveform, verdict) for any user to interpret results easily

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

6	<i>Other</i>	Additional maintainability requirement	Codebase organized into independent, reusable modules (transcription, semantic analysis, audio features, scoring, feedback, reporting) to support easy future feature additions
---	--------------	--	---